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Nốt

One positive note every day

Đoàn Thị Kim Thúy · SE190992

Phùng Thế Quân · SE193519

Phan Đức Duy · SE193591

Phan Hữu Phước · SE161951

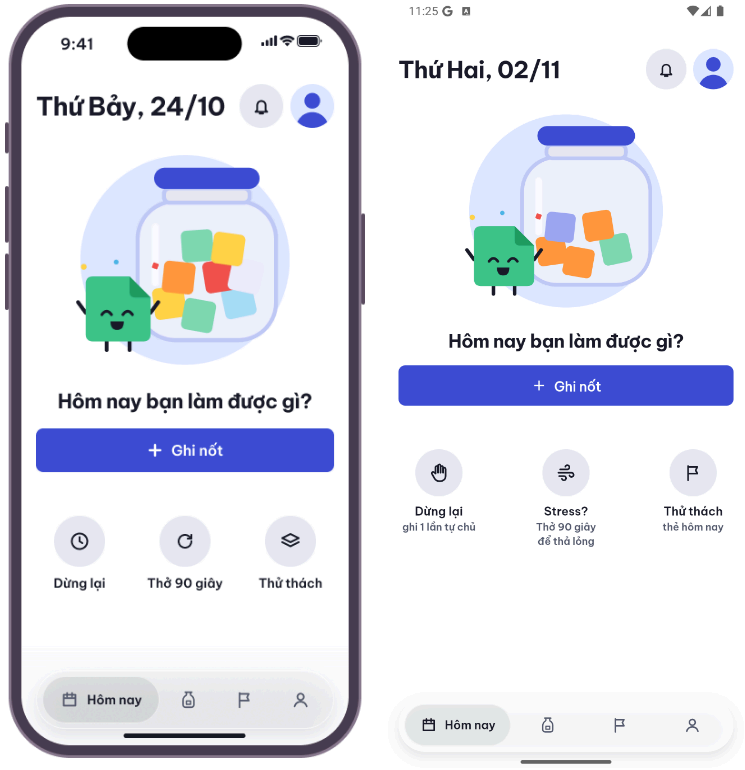


Introduction

Tiny Wins (Nốt) is a mobile app for burned-out final-year students like Minh (22): every evening they drop one small win into a jar instead of staring at an unfinished to-do list. After a rough day, the jar hands back an old win to remember. Small bravery and movement challenges, a one-tap self-control log and a guided journal help them rebuild confidence, one note at a time. Success criterion: log a mood and one win in under 30 seconds.

Figma vs app

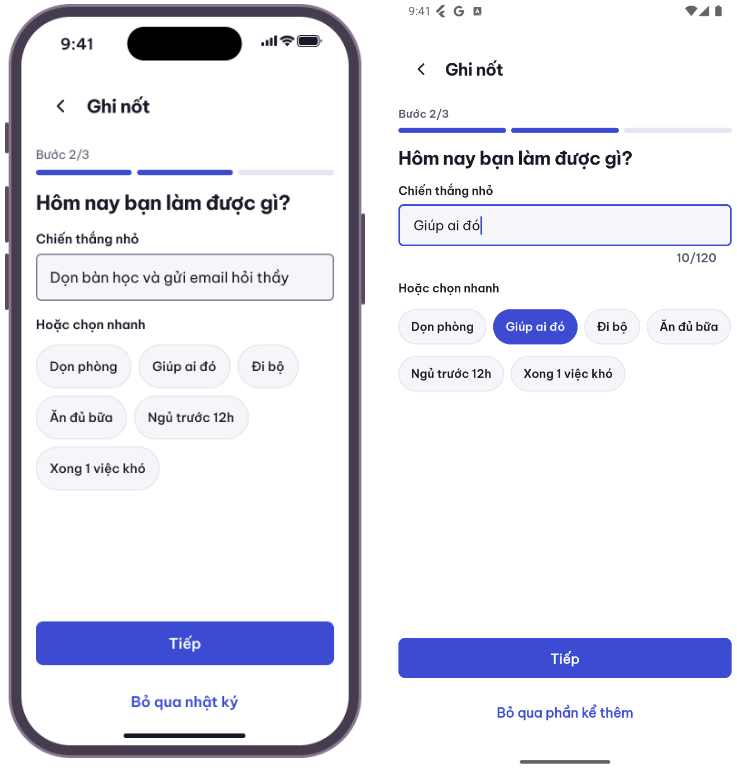
Today



Figma App

- Android status bar (D-04)
- New shortcut icons and captions (D-25)
- "Ghi nốt" asks win or journal first (D-29)

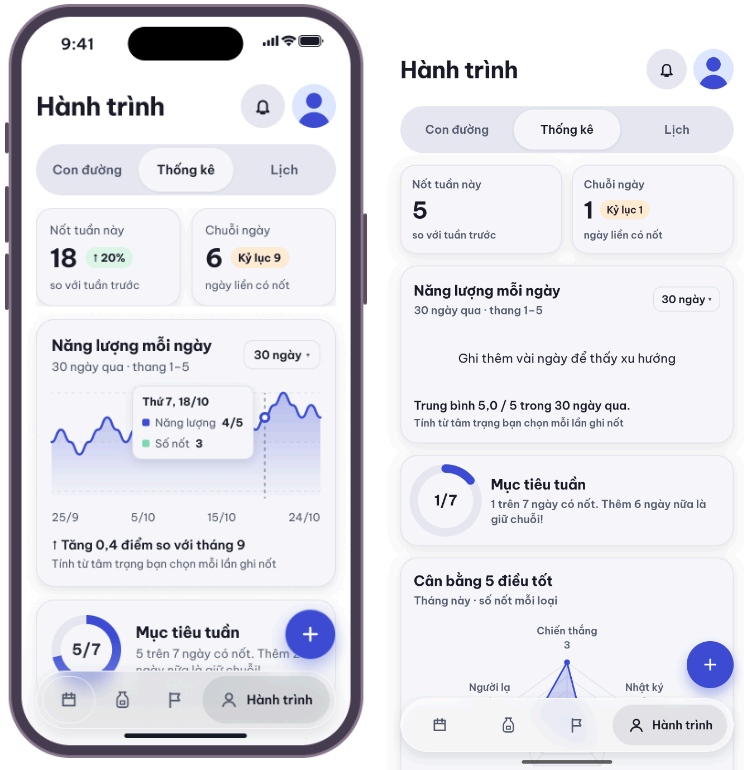
Write a note



Figma App

- "Bỏ qua nhật ký" is now "Bỏ qua phần kể thêm" (D-28)
- 0/120 counter under the field (D-02)
- AI free-tell is planned, not built

Stats

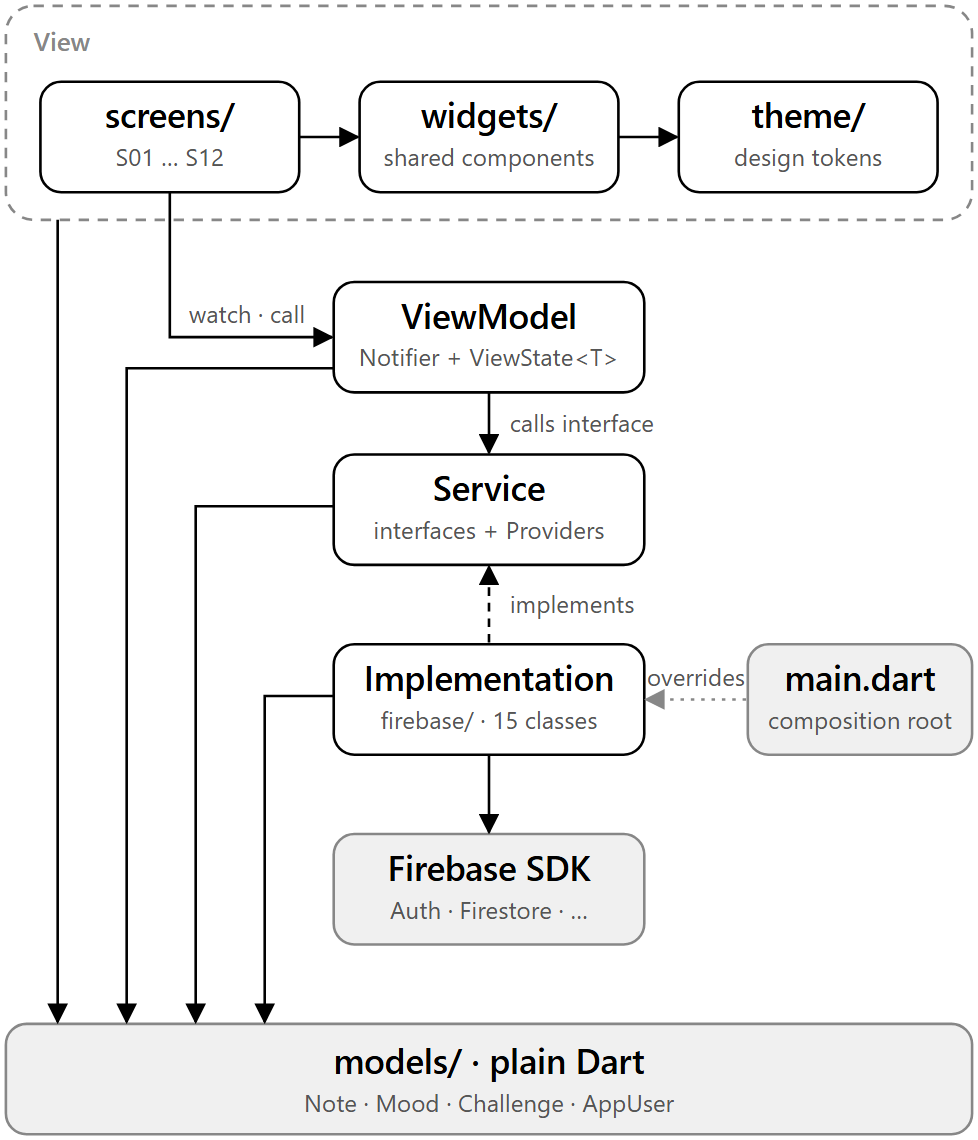


Figma App

- Compared with the previous 30 days (D-18)
- Chart scales to the card width (D-09)
- Weekly goal opens the weekly card (D-06)

MVVM

```
lib/
├── main.dart                  composition root
├── app.dart                   MaterialApp.router
├── router/
│   └── auth_redirect.dart     signed out → /welcome
├── theme/
│   └── app_tokens.dart        spacing, radius
├── models/
│   └── note.dart              plain Dart
├── services/
│   └── note_repository.dart
│                              interface + Provider
├── firebase/
│   └── firestore_note_repository.dart
│                              implementation
├── viewmodels/
│   ├── view_state.dart        Loading | Empty | Failure | Data
│   └── today_view_model.dart
│                              Notifier + ViewState
├── screens/today/
│   └── today_screen.dart      switch on ViewState
├── widgets/
│   └── app_button.dart        shared component
└── utils/
    └── dates.dart             pure helpers
```



Design system

Primary

primary	primary-container	primary-dark
#3D4FD6	#DDE1FF	#2B399F

Surface

surface	surface-card	surface-variant	outline
#F6F7FB	#FFFFFF	#E7E9F3	#7A8094

Text

on-surface	on-surface-variant
#1B1D2A	#555B6E

Status

error	warning	success
#B3261E	#A84F00	#1F9D63

Heat

heat/1	heat/2	heat/3	heat/4	heat/5
#A9DCF5	#7FD9B3	#FFD24A	#FF9A3D	#F0524B

Tint

blue	orange	red	green	violet	yellow	indigo
#E3F3FB	#FFEBD6	#FDE1DF	#DDF5EA	#ECEBFC	#FFF4CC	#E4E7FF

Nốt nhắc bạn

Display · 44 Bold

Nốt nhắc bạn

Headline · 28 Bold

Nốt nhắc bạn

Title · 20 Bold

Nốt nhắc bạn

Body · 16 Regular

Nốt nhắc bạn

Label · 14 SemiBold

Nốt nhắc bạn

Caption · 12 Regular

Space

2 4 8 12 16 24

Radius

4 6 8 12

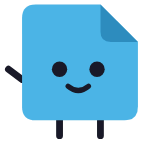
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Meet Tụi Nốt and the jar

Eight mascot moods and one jar that fills up with your wins



Nốt Xong
A win saved · Vui



Nốt Thong Thả
Welcome, breathing · Ổn



Nốt Đợi
Loading, empty · Bình thường



Nốt Ngủ
AI resting · Mệt



Nốt Lạc
Errors · Rối



Nốt Tập Trung
Challenges, self-control



Nốt Vội
Design system only



Nốt Cháy
Design system only

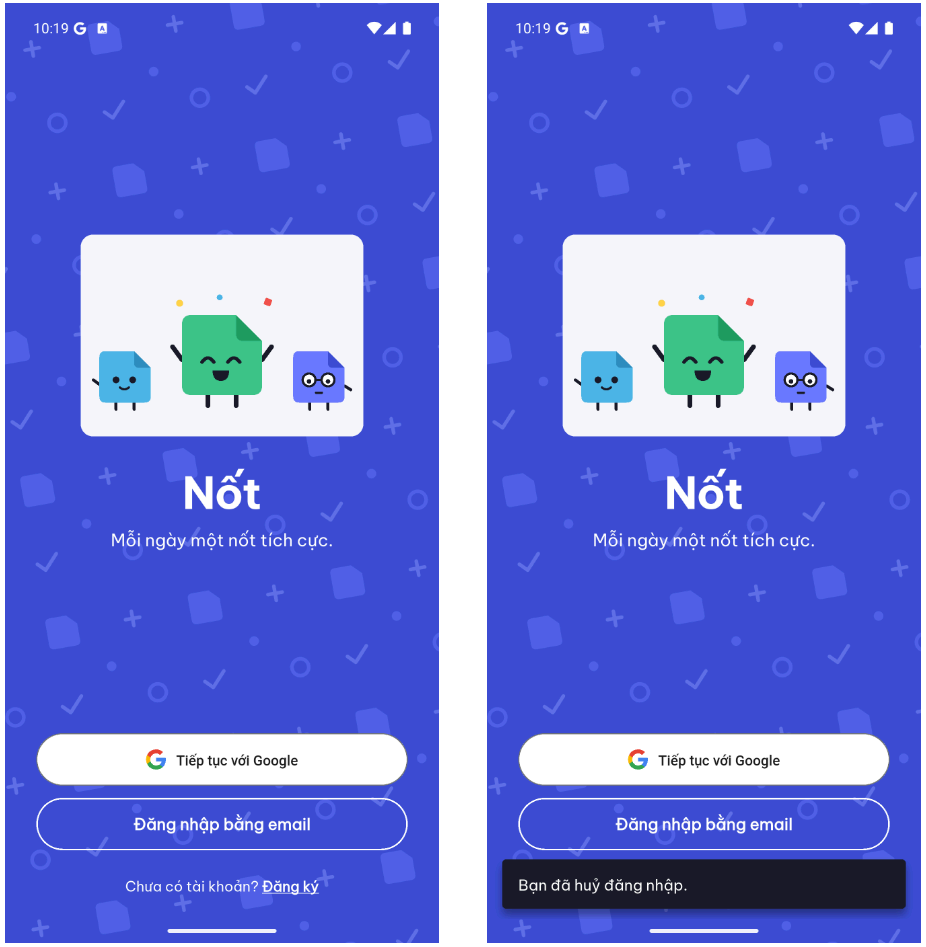


- **Hũ chiến thắng:** every saved note drops in as a slip
- Slip colour shows the type: Chiến thắng, Tự chủ, Thử thách, Nhật ký
- Shows the 8 newest; the number counts this month's notes
- Each mood has its own face, so mood is not shown by colour alone

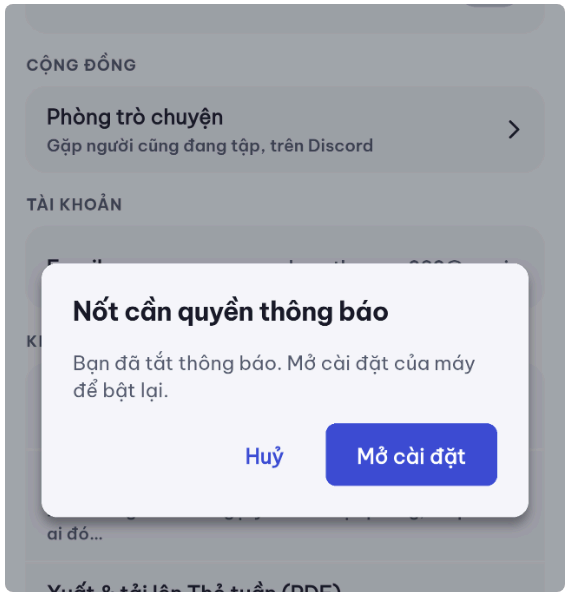
Authentication

```
$ firebase auth:export users.json
Exported 1 account(s) · provider: google.com

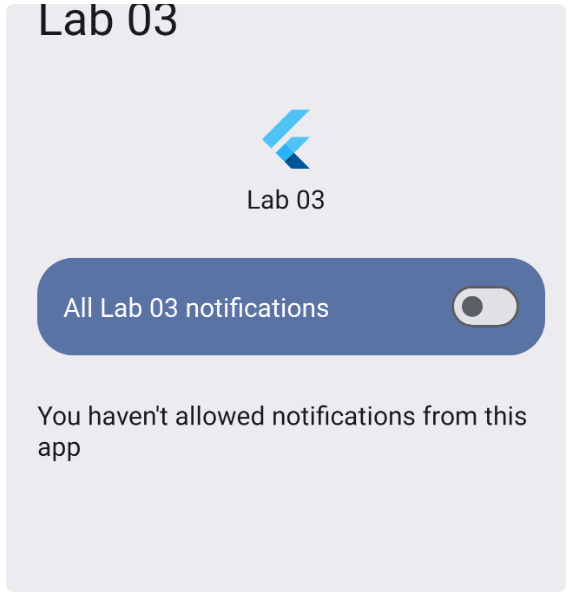
Cancel   → "Bạn đã huỷ đăng nhập."
Offline  → "Không có mạng. Kiểm tra Wi-Fi rồi thử lại."
Sign out → local data cleared, back to /welcome
```



Cloud Messaging



Denied → open settings



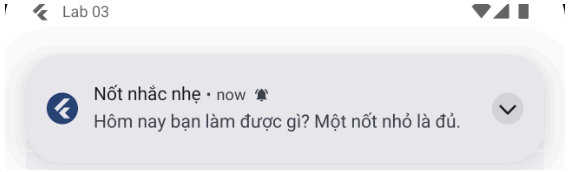
System settings



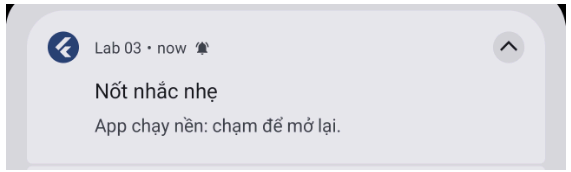
Inbox: 3 states



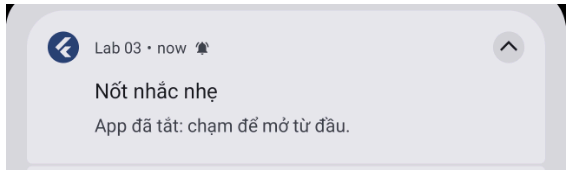
Tap · route=/jar



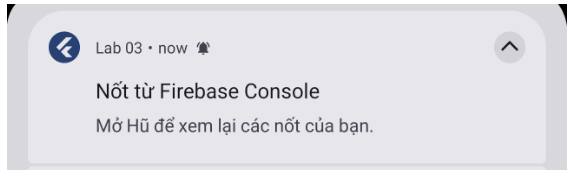
Foreground



Background

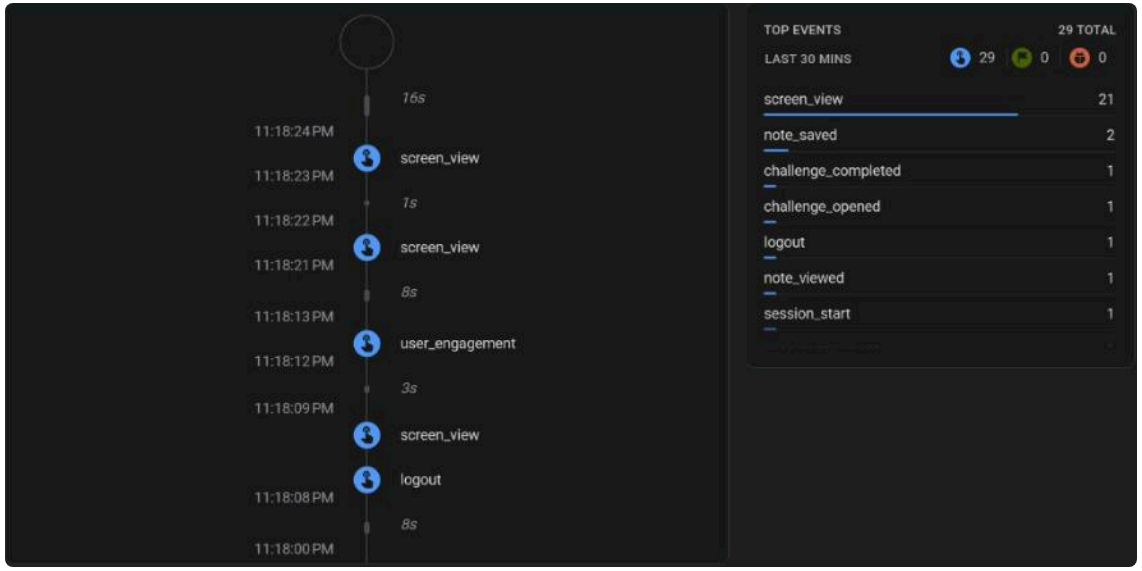


Terminated



Sent from Console

Analytics



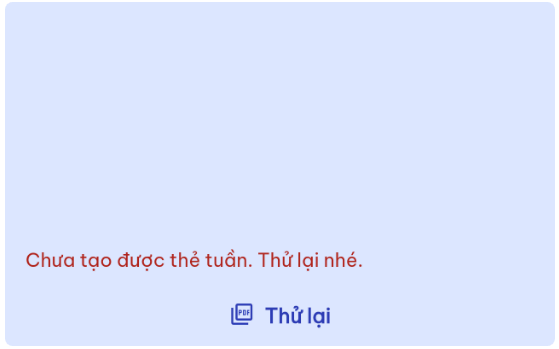
DebugView · top events of one run on the emulator

Eight custom events go through an `AnalyticsService` interface (Firebase in the app, a console logger on the Emulator). No titles, note text, emails or names are logged. The Firebase user ID is set after sign-in and cleared on sign-out.

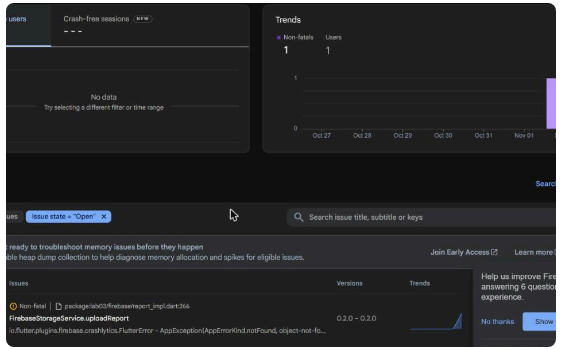
- **login / logout** · `method`
- **note_saved** · `mood, note_type, has_journal, duration_ms`. Tracks the 30-second goal: 7.7 s in this run.
- **note_viewed** · `note_type`
- **challenge_opened / challenge_completed** · `challenge_id, note_type`
- **self_control_logged** · `has_habit`
- **report_exported** · `format`. Fires after a successful upload, so it needs the Storage bucket.

Checked three ways: DebugView (left), `adb logcat` for all events except report_exported, and unit tests that assert the parameters.

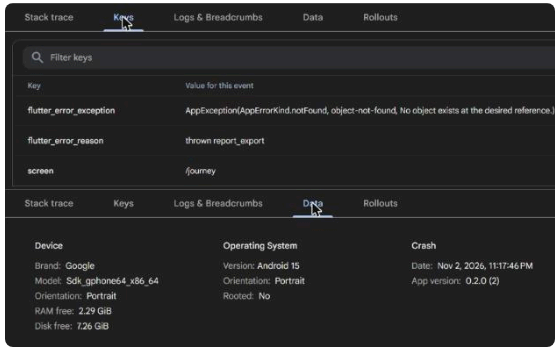
Crashlytics



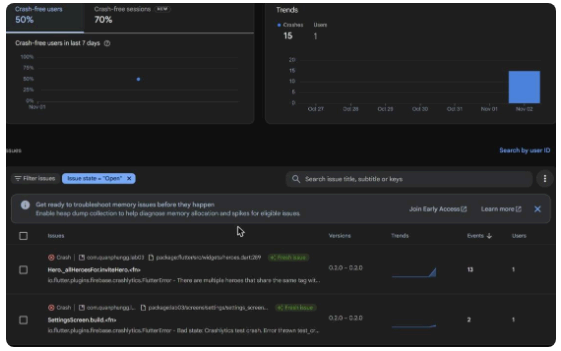
1 · User still sees a message



2 · Non-fatal in Console



3 · Key screen · user ID



4 · Test crash + a real Hero error

Crashlytics collects errors from real devices, so we see problems the Emulator never shows. Two kinds reach the dashboard:

- **Non-fatal**: the app caught the error and kept running. Exporting the weekly card fails because the Storage bucket is not provisioned; the user sees "Chưa tạo được thẻ tuần. Thử lại nhé." and the app records the error (1, 2).
- **Context on every report**: the Firebase user ID and a custom key `screen` (here `/journey`) show who hit it and where (3).
- **Crash**: a dev-only "Thử crash" button sends a test crash. The same dashboard also surfaced a real Hero error, caused by two add-note buttons sharing one tag. We fixed it with `heroTag: null` (4).

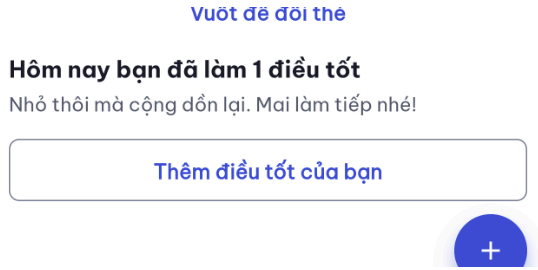
Remote Config



groups_enabled = true



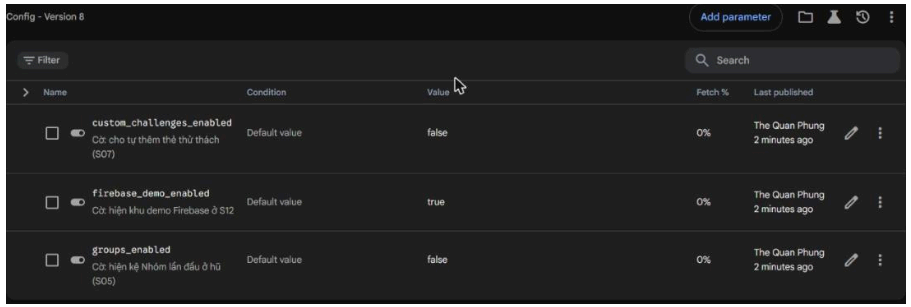
groups_enabled = false



custom_challenges_enabled = true

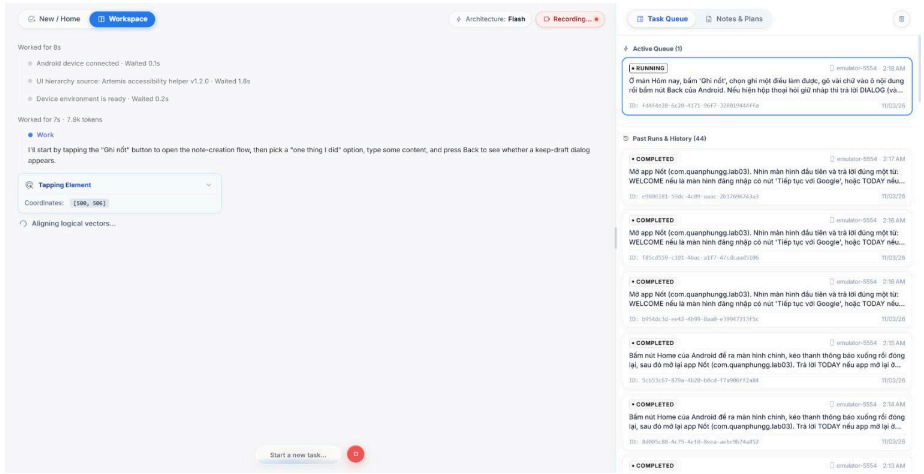


custom_challenges_enabled = false



Console · three feature flags

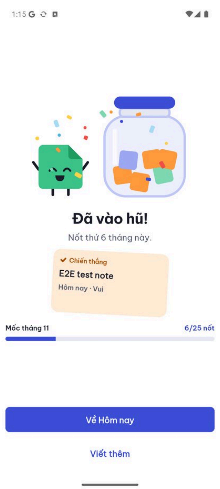
ARTEMIS E2E



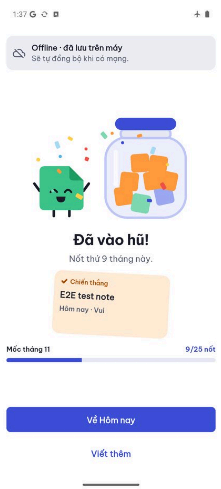
ARTEMIS Workspace · agent steps and run history (44 runs)

An AI agent (`deepseek-v4.1-flash` via OpenRouter) drives the app on the Android 15 emulator over ADB. Each case runs 3 times and asserts on the value the agent reports.

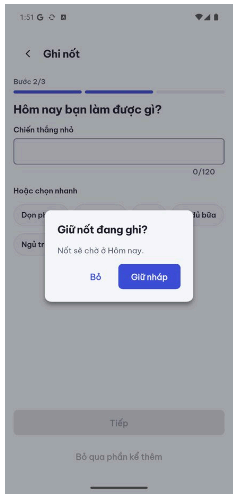
- **12 cases:** F1 save, empty, draft dialog, offline · F2 · F3 · tabs · flags · native Home and relaunch
- **35 of 36 runs pass.** T06 (offline) timed out once.
- First run was 31 of 36: two assertions in our tests were wrong (fixed, re-run 6 of 6).
- Not done yet: the 15 to 30 minute exploratory session.



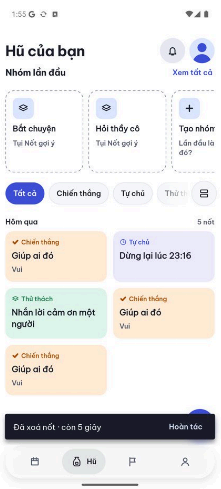
T02 · note saved



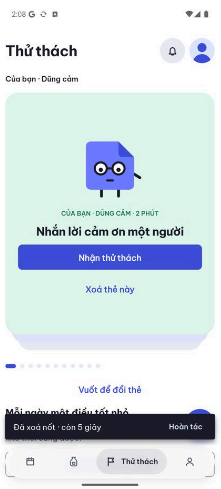
T06 · saved on device



T05 · keep-draft dialog



T03 · note deleted



T09 · challenge opened

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AI code review

CodeRabbit reviewed every PR: 52 findings on #28, #29 and #38

Finding	Kind	Decision	Reason · commit
Restored session has no Analytics user ID	Correctness	Fixed	Router sets the ID on every auth change · 211a4d5
FCM inbox survives sign-out	Privacy · Major	Fixed	Sign-out clears the inbox · 736453a
Foreground notification has no route payload	Correctness	Fixed	Payload carries the route, tap opens it · 5e13651
No context.mounted check after await	Stability	Accepted	Settings can throw if the user leaves mid-call
Pass iOS settings to notifications init	Correctness · Major	Won't fix	The app ships on Android and web only
Add-card sheet can submit twice	Correctness	Rejected	canSubmit already reads _saving

51 of 52 right · 40 real bugs, 11 not worth it, 1 wrong · **3 fixed, one commit each** · full list in docs/code-review.md

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Lessons learned

- **Test on the real project early.** The Emulator has no Analytics, Crashlytics, Remote Config or FCM, so Stage 5 only worked once we connected the real project.
- **Design documents can disagree.** Two Lab 2 documents set different title limits; we picked one and logged it (D-11).
- **Labels that users misread must change.** "Bỏ qua nhật ký" became "Bỏ qua phần kể thêm" (D-28).
- **An AI tester needs checking too.** The first ARTEMIS run failed 5 of 36; two causes were wrong assertions in our tests, not app bugs.
- **AI review only helps if someone reads it.** CodeRabbit was right on 51 of 52 findings, but we had fixed only 1 of 40 real bugs when we merged.